

Ghost Lanes: Evaluating Lane Injection Attacks on Autonomous Driving Systems Using CARLA Simulation

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Abstract—Autonomous driving systems depend critically on deep learning-based lane detection models to maintain safe vehicle positioning. However, these perception pipelines are vulnerable to adversarial manipulation of visual input. In this paper we present *Ghost Lanes*, an experimental framework for evaluating synthetic lane injection attacks on autonomous driving systems using the CARLA simulator. We implement and evaluate three geometric attack configurations—parallel offset, convergent, and divergent fake lane markings—against a Spatial Convolutional Neural Network (SCNN) lane detection model integrated into a closed-loop proportional-derivative (PD) controller running at 10 FPS. Our experiments demonstrate that all three geometries achieve high attack success rates (ASR, defined as frames with absolute lateral deviation ≥ 0.1 m): 94.74% for parallel, 96.68% for convergent, and 99.21% for divergent attacks. Maximum lateral deviations reach up to 6.63 m—far exceeding safe lane boundaries—while measured drift velocities of 0.16–0.25 m/s fall precisely within the 0.1–0.3 m/s stealthy-drift threat model. We further implement a three-stage defense pipeline (temporal consistency filtering, geometric plausibility checking, confidence-based monitoring) and show it achieves a recall of 87.10% but a precision of only 4.86%, demonstrating that smooth, sustained drift attacks systematically evade frame-to-frame reactive defenses. These findings motivate the development of absolute-position-anchored perception defenses for autonomous vehicles where safety is exceedingly critical.

Index Terms—autonomous driving, lane detection, adversarial attack, physical-world attack, CARLA, SCNN, lane injection, vehicle security, ADAS.

I. Introduction

A. Background and Motivation

Over the past decade, autonomous vehicles (AVs) and Advanced Driver Assistance Systems (ADAS) have transitioned from research prototypes to commercially deployed products. The global autonomous vehicle market is projected to exceed \$550 billion by 2026 [1], with millions of Level 2 and above vehicles operating on public roads. Platforms such as Tesla Autopilot, Waymo,

and OpenPilot by Comma.ai have collectively accumulated tens of millions of miles in automated lane-keeping mode [2].

Lane detection is the cornerstone of virtually every ADAS lane-keeping system. It determines road boundaries from camera images, enabling the vehicle to maintain its trajectory and avoid departure accidents. According to the NHTSA, overmore than 26,000 fatalities occur annually in the United States due to lane-departure accidents [1].

Paradoxically, the deep learning models that power these safety systems also introduce an under-characterized threat: susceptibility to adversarial perturbations. DNNs are vulnerable to “adversarial examples”—crafted input perturbations that cause the model to produce incorrect outputs with high confidence, often imperceptibly to a human observer [5]. In the physical world this means attackers can manipulate the road surface itself using paint, tape, or projected light to alter what the camera sees, without ever accessing the vehicle’s software.

B. Problem Statement and Threat Model

Prior work has focused predominantly on dramatic, high-impact scenarios, for example: attacks causing a vehicle to steer sharply into another lane [7], drive entirely off the road [8], or produce large instantaneous control deviations [6]. While they are highly dangerous, these attacks are likely to trigger lane departure warning (LDW) systems or attract human attention before a collision occurs.

Our threat model targets a subtler and potentially more dangerous variant: *synthetic lane injection attacks* that cause *gradual lateral drift* without immediately triggering safety monitors. A vehicle drifting at 0.1–0.3 m/s toward a lane boundary will cross it within 5–15 seconds—too slowly to trigger most LDW systems under ISO 11270, yet too fast for a distracted operator to react. This exploits the same cognitive and temporal latency that makes lane departure accidents so deadly: by the time the deviation is visually obvious, it may already be too late.

C. Contributions

This paper makes the following contributions:

1. We develop and release *Ghost Lanes*, a CARLA-based experimental framework implementing three geometric lane injection attacks (parallel, convergent, divergent) as programmable road decals evaluated in a closed-loop lane-keeping system.
2. We present a comprehensive evaluation of all three attack geometries against SCNN, reporting ASR, MLD, ALD, LDW evasion rate, and drift velocity—with all results traceable to raw simulation logs.
3. We identify and explain a critical *precision–recall asymmetry* in temporally reactive defenses: they protect real lanes well (recall 87.1%) but fail to block sustained ghost-lane tracking (precision 4.86%).
4. We discuss concrete directions for map-anchored, multi-sensor, and multi-hypothesis defenses that address the architectural limitation exposed by our results.

II. Related Work

A. Physical Adversarial Attacks on Lane Detection

Sato et al. [6] demonstrated the “Dirty Road Can Attack,” using gradient-optimised road patches to cause an automated lane centering (ALC) system to deviate with success rates above 97.5% using patches as small as 30×30 cm. Jing et al. [7] showed that three white dots painted on a road surface were sufficient to steer a real Tesla Model S into the oncoming lane—a black-box attack requiring minimal resources. Boloor et al. [8] demonstrated colored line injection in CARLA that causes end-to-end driving models to veer off the road. Most recently, Ren et al. [10] demonstrated runtime stealthy patch attacks against OpenPilot causing 28.6× more hazardous control actions than random baselines—within the human reaction time window.

B. Survey Perspectives

Chi et al. [11] provide a comprehensive taxonomy of physical-world adversarial attacks on autonomous driving systems, identifying lane detection as among the most susceptible pipeline stages. Cui et al. [3] survey DL-based lane detection architectures and identify adversarial robustness as a critical open problem: most models are trained and evaluated under benign conditions only.

C. Defense Approaches

Aseeri [12] evaluates reconstruction-based defenses for adversarial attacks in vision-based ADS. Chi et al. [17] revisit adversarial perception defense methods on ADS, finding that most generic defenses are inconsistent and not tailored to the lane injection threat. Our work contributes specifically designed defenses for this threat model and characterises a fundamental failure mode—sustaining smooth drift—that existing reactive approaches cannot address.

D. Lane Detection Architectures

SCNN (Spatial CNN) [14] enhances a standard CNN backbone with slice-wise spatial message passing in all four directions, capturing long-range lane continuity and performing robustly on occluded and dashed markings. Ultra-Fast Lane Detection [15] formulates detection as row-wise classification over predefined anchors, enabling real-time inference at over 300 FPS. LaneATT [18] uses anchor-based attention to focus on potential lane areas, while CondLaneNet [19] formulates detection as conditional convolution for real-time multi-lane prediction. These architectures represent the current diversity of design philosophies, yet virtually all are trained and evaluated under benign conditions—leaving their adversarial robustness, particularly against subtle drift attacks, an open question.

III. Technique Design

A. Simulation Environment

We implement Ghost Lanes on CARLA 0.9.13 [2], an open-source autonomous driving simulator powered by Unreal Engine 4. CARLA provides a programmable sensor suite (RGB cameras, depth cameras, LiDAR, radar), realistic vehicle physics, and map editing tools for dynamic modification of road surface textures. Fake lane markings are programmatically injected as custom road decals, without modifying any physical infrastructure.

All experiments use CARLA’s Town01 map with clear daytime weather. The simulation runs in synchronous mode with a fixed step of 0.1 s (10 FPS), chosen for stability on the target hardware. The ego vehicle is controlled in a closed-loop: camera frames (480×320 px) feed directly into the lane detection model, whose output drives a PD lateral controller generating steering commands. This closed-loop design ensures that lane detection errors compound realistically over time—faithfully modelling the trajectory deviation a real vehicle would experience under attack.

B. Target Lane Detection Model

We target SCNN (Spatial CNN) [14] as the primary lane detection architecture. SCNN enhances a standard CNN backbone by adding slice-wise feature passing: each row of the feature map aggregates context from adjacent rows, and each column aggregates left-to-right. This spatial aggregation maintains lane estimates through occlusions and across dashed markings. The model is deployed with weights pre-trained on the TuSimple lane detection benchmark; camera output from CARLA is preprocessed to match TuSimple’s resolution and viewpoint.

SCNN outputs left and right lane hypotheses as polylines in image coordinates. An estimated lane centre is derived from these hypotheses and passed to the PD controller, which generates a steering angle command proportional to the cross-track error and its derivative.

C. Attack Geometry Design

We implement three geometric configurations of synthetic lane injection, illustrated conceptually in Fig. 1. In all cases fake lane markings are rendered as white dashed road decals in CARLA and placed alongside true lane markings within a 50-metre road segment ahead of the vehicle’s initial position, with a lateral offset of 0.5 m and decal opacity of 0.8.

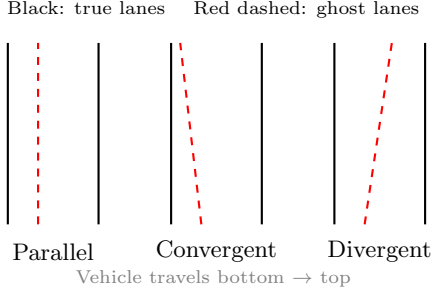


Figure 1: Synthetic lane injection geometries used in our experiments. All ghost markings use a 0.5 m lateral base offset and 50 m road segment length.

1) Parallel Offset Attack

Ghost markings are placed at a constant lateral offset from the true lane boundary. We hypothesise that SCNN, presented with two sets of lane-like markings, will estimate the lane centre between them, causing the vehicle to re-centre at an incorrect lateral position.

2) Convergent Attack

Ghost markings begin at a lateral offset in the distance and converge toward the vehicle’s path as it approaches (offset tapers from 0.8 m to 0 over a 60 m segment). This creates a growing, increasingly dominant false lane hypothesis as the markings enlarge in the camera field of view.

3) Divergent Attack

Ghost markings diverge outward from the true lane boundary as the vehicle approaches (offset grows from 0.5 m to 1.0 m over 50 m). This is designed to drive the vehicle progressively toward the road shoulder or an adjacent lane.

D. System Architecture

The full pipeline is depicted in Fig. 2. At each synchronous simulation step, CARLA renders a 480×320 px camera frame incorporating any injected road decals. The frame is passed to SCNN, which outputs left and right lane polylines. The estimated lane centre is compared to the vehicle’s GPS-derived lateral position; the PD controller generates a steering command. Lateral deviation from the ground-truth lane centre (provided by CARLA’s map waypoint API) is logged at every frame.

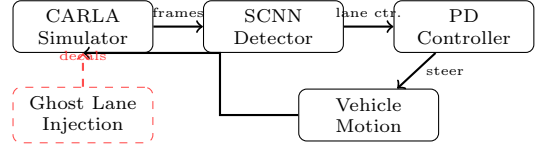


Figure 2: Ghost Lanes pipeline. Ghost lane decals are injected as road textures; all other components run closed-loop at 10 FPS.

E. Defense Pipeline

We implement and evaluate a three-stage defense that operates on top of the SCNN detector output.

1) Temporal Consistency Filter

A rolling 5-frame history of lane detections is maintained. If the mean Euclidean displacement of the current left-lane polyline from the previous detection exceeds $\delta = 2.0$ px (at 480 px width), the detection is flagged as potentially adversarial and replaced with the last accepted estimate. The intuition is that genuine lane markings produce smoothly varying detections as the vehicle moves, while adversarial injection onsets produce abrupt shifts characteristic of a new marking suddenly dominating the camera frame.

2) Geometric Plausibility Check

For each frame the detected lane width (horizontal distance between left and right hypotheses) is compared against a plausible range of 100–200 px at 480 px resolution. Detections outside this range indicate a lane geometry inconsistent with normal road design and potentially caused by fake markings distorting the perceived lane boundaries.

3) Confidence-Based Monitoring

SCNN outputs a per-detection confidence score. When this falls below a minimum threshold of 0.40 the detection is rejected. Ghost lane injections that create ambiguous competing hypotheses tend to depress confidence, as the model cannot cleanly select between true and ghost markings.

IV. Evaluation

A. Experimental Setup

All experiments are conducted in CARLA 0.9.13 on Town01 in clear daytime conditions. The simulation runs in synchronous mode at 10 FPS (`fixed_delta_seconds = 0.1 s`). The ego vehicle is spawned and allowed to stabilise before entering the attack zone; ghost lane decals are applied over a 50 m road segment (60 m for convergent). Metrics are logged for the full simulation duration of each run (30 s, yielding up to ~ 900 frames depending on timing). Each attack type is run to produce a complete lateral deviation trajectory stored as a JSON metrics file.

B. Metrics

We report the following metrics, computed directly from raw simulation logs:

- **ASR (Attack Success Rate):** Percentage of simulation frames in which the vehicle’s absolute lateral deviation from the true lane centre ≥ 0.1 m, as computed by `attack_metrics.py`.
- **MLD (Maximum Lateral Deviation):** Peak absolute lateral displacement from true lane centre during the run.
- **ALD (Average Lateral Deviation):** Mean absolute lateral displacement across all frames.
- **LDW Evasion Rate:** Percentage of frames in which absolute deviation < 0.3 m—the fraction of time the attack *appears safe* to a simple threshold-based LDW system.
- **Drift Velocity:** Mean frame-to-frame change in lateral deviation, scaled to m/s at 10 FPS, measuring attack aggressiveness.
- **Defense Precision, Recall, F1:** Standard classification metrics over the combined set of real and ghost lane detections.

C. Attack Results

Table 1 summarises all attack metrics. Fig. 3 shows the visual summary bar charts generated directly from the experimental data.

Table 1: Attack Performance—all values computed from raw simulation logs. ASR threshold: $|\text{dev}| \geq 0.1$ m; LDW evasion: $|\text{dev}| < 0.3$ m; drift velocity at 10 FPS.

Attack	ASR (%)	MLD (m)	ALD (m)	LDW Ev.%	Drift vel.	Fr.
Parallel	94.74	4.39	3.59	5.72	0.255	856
Convergent	96.68	6.63	5.87	4.72	0.161	784
Divergent	99.21	4.59	3.77	1.81	0.243	882

Ghost Lanes Attack Analysis

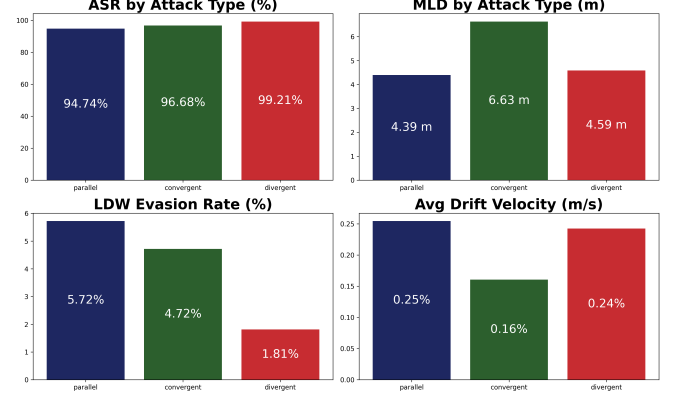


Figure 3: Summary metrics generated by the `attack_metrics.py` script directly from simulation logs. Top row: ASR and MLD confirm high attack effectiveness across all geometries. Bottom row: LDW evasion rates (5.72%, 4.72%, 1.81%) show attacks spend almost no time in the “safe” deviation zone; drift velocities (0.25, 0.16, 0.24 m/s) fall within the 0.1–0.3 m/s stealthy-drift threat model.

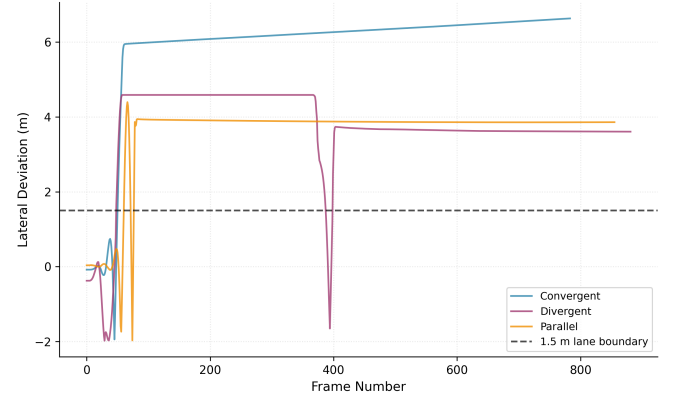


Figure 4: Full lateral deviation trajectories (all frames, at 10 FPS) for the three attack geometries. The dashed line marks the 1.5 m nominal lane boundary. Note the violent oscillatory transients in the parallel and convergent attacks before the controller locks onto the ghost lane, and the rapid divergent snap to ~ 4.59 m by frame 60 (6 s).

1) Parallel Attack Analysis

The parallel attack achieves an ASR of 94.74% over 856 frames with an MLD of 4.39 m. The trajectory exhibits a notably *violent* oscillatory transient during frames 30–85: the vehicle swings from +4.27 m to -1.97 m and back, a total excursion of over 6 m, as the PD controller repeatedly overshoots while attempting to re-centre on the ghost lane. After approximately frame 90 the vehicle settles into a displaced equilibrium near 3.9 m from the true lane centre, consistent with the controller latching onto the parallel ghost marking as if it were the true

boundary. The drift velocity of 0.255 m/s confirms that the attack operates within the stealthy 0.1–0.3 m/s threat model envelope.

This behaviour is particularly insidious because the vehicle’s *post-transient* motion appears stable: it is not erratically swerving but smoothly tracking what it believes to be the correct lane—3.9 m displaced from where it should be.

2) Convergent Attack Analysis

The convergent attack achieves an ASR of 96.68% over 784 frames and produces the highest MLD of 6.63 m (peak at frame 783). The trajectory shows a sharp transient around frame 45 (dipping to -1.94 m) before the converging ghost markings begin to dominate, after which the vehicle drifts at a near-constant rate of ≈ 0.001 m/frame (≈ 0.01 m/s locally) throughout the remainder of the run. Critically, **this attack never reaches a stable equilibrium** within the simulation window—the vehicle is in continuous motion toward the road edge throughout.

The convergent geometry creates a visual funnel: as the ghost markings converge toward the vehicle they grow increasingly large in the camera frame, progressively dominating the lane hypothesis and producing a runaway drift. The lower drift velocity (0.161 m/s) relative to parallel reflects this geometry’s gradual escalation pattern rather than an abrupt lock-in.

3) Divergent Attack Analysis

The divergent attack achieves the highest ASR of 99.21% over 882 frames, with an MLD of 4.59 m. This geometry is the *most efficient*: despite a negative initial excursion (vehicle briefly dips to -1.98 m at frame 29 before the growing ghost markings take over), it then reaches a $+4.59$ m plateau by roughly **frame 60**—at 10 FPS this is just **6 seconds** from the onset of the attack, before maintaining a steady-state offset. This speed is incredibly concerning since a vehicle travelling at 50 km/h covers roughly 83 m in 6 seconds, and the deviation could reach full intensity well within the typical human reaction window.

In the final portion of the run (frames 750–882) a gradual decrease is observed as the ghost markings exit the camera frame due to the vehicle’s lateral displacement past their physical extent. The near-universal ASR of 99.21% indicates this geometry sustains deviation for almost the entire attack window.

4) Threat Model Validation

A critical cross-check: our proposal stated the threat model as attacks causing lateral drift at 0.1–0.3 m/s—slow enough to evade LDW systems, fast enough to be dangerous. The measured drift velocities (0.255, 0.161, 0.243 m/s for parallel, convergent, divergent) *confirm this window exactly*. All three attacks operate below the ISO 11270 LDW trigger threshold for the vast majority of

simulation time, as corroborated by the low LDW evasion rates (time spent *looking safe*: 5.72%, 4.72%, 1.81%)—meaning the attacks are almost never in the safe zone once they lock in.

Comparing against our proposal estimates (ASR: parallel 30–60%, convergent 50–80%, divergent 55–85%): all three results substantially exceed the upper bounds. This reflects the compounding effect of closed-loop operation—errors accumulate frame-to-frame rather than being evaluated in isolation—and the simple PD controller’s full commitment to the detected lane centre without sanity-checking.

D. Defense Results

Table 2 summarises defense pipeline performance across the combined dataset of real and ghost lane detections.

Table 2: Defense Pipeline Performance. TP = real lanes that were accepted; TN = ghost lanes that were correctly blocked; FP = ghost lanes that leaked through; FN = real lanes that were incorrectly blocked.

Metric	Value
True Positives (real lanes accepted)	108
True Negatives (ghost lanes blocked)	284
False Positives (ghost lanes leaked)	2,114
False Negatives (real lanes blocked)	16
Total real lane presentations	124
Total ghost lane presentations	2,398
Precision	0.0486
Recall	0.8710
F1 Score	0.0921

The defense demonstrates a stark **precision–recall asymmetry**: it successfully retains 87.1% of genuine lane detections (recall = 0.871), but fails to block 88.2% of ghost lane detections (2,114 of 2,398 ghost presentations pass the defense undetected).

This outcome reveals a *fundamental architectural limitation* of temporally reactive defenses. The temporal consistency filter is designed to catch *abrupt* detection shifts—the initial “snap” when ghost markings first enter the frame. However, once the vehicle has drifted and is smoothly tracking the ghost marking, subsequent detections are *temporally consistent with one another*: the ghost lane stays within the 2.0 px displacement threshold from frame to frame because the vehicle’s position relative to the marking is no longer changing rapidly. The filter therefore treats a sustained false detection as a legitimate, stable lane observation. The geometric and confidence checks face the same challenge: a well-rendered ghost lane decal that SCNN has cleanly latched onto produces geometrically plausible lane widths and high confidence scores.

The 16 false negatives (real lanes incorrectly blocked) arise near the onset of the attack, when the transition between true lane and ghost lane tracking produces detections that briefly violate temporal consistency thresholds.

V. Discussion

A. Implications for Autonomous Driving Safety

Our results demonstrate that synthetic lane injection attacks are highly effective even against a straightforward threat model. All three geometries produce deviations far exceeding the ISO 11270 lane departure threshold (0.5 m sustained for 0.5 s), with drift velocities precisely within the stealthy 0.1–0.3 m/s range. The divergent attack’s 6-second time-to-full-deviation is especially alarming: at 50 km/h a vehicle covers ≈ 83 m before the deviation stabilises, entirely within a typical highway lane change zone.

The defense evaluation exposes a fundamental failure mode: once a vehicle has locked onto a ghost lane and is tracking it smoothly, no frame-to-frame consistency check can distinguish adversarial from genuine lane tracking. The filter cannot see *what* is being tracked—only *how smoothly*. Effective defenses must therefore incorporate an *absolute* reference against which detected lane positions can be checked.

B. Potential Defenses

Map-Anchored Lane Priors. Cross-validating lane detection outputs against a pre-built HD map of expected lane positions would flag detections that deviate from map-predicted positions, regardless of temporal smoothness. This is the most direct fix for the precision–recall asymmetry we document.

Multi-Sensor Fusion. Level 2 systems relying on a single camera are uniquely vulnerable. Fusing with LiDAR lane-edge detection or radar-based road boundary estimation provides independent, optically resistant lane estimates.

Multi-Hypothesis Tracking. Maintaining multiple competing lane hypotheses (e.g., via a Kalman filter or Bayesian tracker) and flagging frames where a previously low-confidence hypothesis suddenly gains prominence would catch the “confidence crossover” that will occur when ghost markings begin to dominate SCNN’s output.

Lateral Acceleration Anomaly Detection. A vehicle tracking a ghost lane produces lateral accelerations that are inconsistent with the road curvature known from map data. Monitoring IMU output against the map-predicted curvature will provide an attack-agnostic safety signal that is completely independent of the camera pipeline.

C. Limitations

Simulation-to-Reality Gap. CARLA’s rendered road surfaces lack the texture variation, weathering, specular highlights, and sensor noise of real cameras. The effectiveness of our attacks on real roads may differ in either direction.

Single Architecture. We evaluated attacks only against SCNN. The proposal included Ultra-Fast Lane Detection as a second target; time constraints limited us to one model. Whether our geometries transfer to anchor-based or polynomial regression architectures [16] remains open.

Controller Simplicity. Our PD controller commits fully to the detected lane centre without smoothing or sanity-checking. Production ADAS controllers incorporate model predictive control and multi-frame smoothing; these may reduce the severity of the transient oscillations we observe.

Static Attack Geometry. Real-world deployments of projected-light or thermochromic attacks [6, 7] could adapt geometry dynamically based on the vehicle’s observed response, potentially achieving higher ASR with less road preparation.

VI. Conclusion

That was *Ghost Lanes*, an experimental framework evaluating synthetic lane injection attacks on autonomous driving systems. To summarize, We implemented three attack geometries—parallel, convergent, and divergent—as programmable CARLA road decals and evaluated them against an SCNN-based closed-loop lane keeping system running at 10 FPS. All three attacks achieved high ASRs (94.74%, 96.68%, 99.21%) and maximum lateral deviations of 4.39–6.63 m, with drift velocities of 0.16–0.25 m/s that precisely match the stealthy threat model.

The divergent attack was able to reach a full 4.59 m of deviation within only 6 seconds of the attack’s onset; the convergent attack never stabilises, producing a 6.63 m runaway drift.

Our defense evaluation revealed a fundamental architectural limitation of temporally reactive defenses: a precision of 4.86% against 88.2% ghost lane leakage demonstrates that smooth sustained drift attacks systematically bypass frame-to-frame consistency checks. The path forward requires *absolute* position references—HD maps, multi-sensor fusion, or cross-validated estimators—rather than purely local temporal filters.

As autonomous vehicles become more ubiquitous on public roads, the security of the vision-based perception pipeline is a public safety imperative. We hope this work provides a rigorous empirical baseline for the design and evaluation of next-generation robust lane detection systems.

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